

longer need to hire and train judicious, highly educated triage professionals. A sufficiently good decision tree could even make it possible to replace human triage professionals with robots, though whether that's a good goal is debatable. A good decision tree may even lead to better decisions than the average human would make, since it could potentially eliminate the unconscious biases that we fallible humans possess. (And in fact, this has already happened: in 1996 and 2002, separate teams of researchers published papers about their success improving triage results for patients complaining of chest pain by using decision trees.)

The branching decision steps described in a decision tree constitute an algorithm. Executing such an algorithm is very simple: just decide which of the two branches you should be on at every node, and follow the branches to the end. But don't obey the suggestions of every decision tree you encounter. Remember that anyone can make a decision tree that prescribes any conceivable decision process, even if it leads to wrong decisions. The hard part of decision trees is not executing the decision tree algorithm but designing the decision tree so that it leads to the best possible decisions. Creating an optimal decision tree is an application of machine learning, though merely following a decision tree is not. Let's discuss the algorithm that creates an optimal decision tree—an algorithm to generate an algorithm—and proceed through the steps of the process to generate an accurate decision tree.

Building a Decision Tree

Let's build a decision tree that uses information about a person to predict how happy they are. Finding the secret of happiness has preoccupied millions of people for millennia, and social science researchers today spill plenty of ink (and burn through plenty of research grants) pursuing the answers. If we had a decision tree that could use a few pieces of information and reliably predict how happy a person is, it would give us important clues about what determines a person's happiness, and maybe even some ideas about how to achieve it ourselves. By the end of this chapter, you'll know how to build such a decision tree.